

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Science Olympiad Worksheet 48

Level 3 — Advanced · Mixed Topics

Class	Class 4 (Primary · Stage 2)	Subject	Science
Level	Level 3 — Advanced	Topic Focus	Mixed Revision (all strands)
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	48 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

This is Worksheet 48 in the 100-part Class 4 Science Olympiad series, in the Level 3 (Advanced) band. These questions go past simple recall: the child must explain why things happen, predict what will occur next, and reason from clues — the style of thinking tested in the NSO and similar science olympiads. Topics are mixed across plants, animals, the human body, matter, force and energy, the environment, and the Earth and universe, with one reasoning question at the end.

Learning Objectives

- Explain how plants make food and how living things depend on one another in a food chain.
- Identify adaptations of animals and classify them by what they eat and how they live.
- Describe changes of state of matter (melting, evaporation) and reason about conductors and magnets.
- Apply ideas of force, friction, energy sources, and the human body's needs and digestion.
- Reason about Earth and Moon, healthy food, and solve a science analogy.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

Q1. A potato is a part of a plant that we eat. Which part of the plant is a potato, and why does it look swollen?

- (A) A root, because it absorbs water
- (B) A stem, because it stores food underground
- (C) A fruit, because it has seeds inside
- (D) A leaf, because it makes food

Q2. Green plants make their own food using sunlight, water and a gas from the air. Which gas do they take in to make food, and which gas do they give out?

- (A) Take in oxygen, give out carbon dioxide
- (B) Take in carbon dioxide, give out oxygen
- (C) Take in nitrogen, give out oxygen
- (D) Take in water vapour, give out carbon dioxide

Q3. A camel can live in the desert for many days without drinking water. Which feature helps it most to survive the hot, sandy desert?

- (A) Its hump stores water directly
- (B) Its hump stores fat, which gives energy and water when needed
- (C) It has gills to breathe under sand
- (D) It has thick fur to keep cool

Q4. The food we eat is broken down so the body can use it. In which order does food travel through the digestive system?

- (A) Mouth → stomach → food pipe → small intestine
- (B) Mouth → food pipe → stomach → small intestine → large intestine
- (C) Stomach → mouth → small intestine → large intestine
- (D) Mouth → small intestine → stomach → food pipe

Q5. Ravi mixes some sugar into a glass of warm water and stirs. The sugar seems to disappear. What has happened to the sugar?

- (A) The sugar has been destroyed
- (B) The sugar has dissolved and is still present in the water
- (C) The sugar has turned into water
- (D) The sugar has changed into a gas

Q6. Three materials are tested: an iron nail, a plastic ruler and a copper wire. Electricity passes easily through two of them. Which two?

- (A) Iron nail and plastic ruler
- (B) Plastic ruler and copper wire
- (C) Iron nail and copper wire
- (D) Only the copper wire

Q7. A ball is rolling on the ground. After some time it slows down and stops, even though no one touches it. Which force makes it stop?

- (A) Magnetic force
- (B) Friction
- (C) Gravity pulling it forward
- (D) Muscular force

Q8. Which of the following is the best example of a renewable source of energy that will not run out?

- (A) Coal
- (B) Petrol
- (C) Sunlight
- (D) Diesel

Q9. A food chain in a garden is: grass → grasshopper → frog → snake. If all the frogs suddenly disappear, what is the MOST likely result?

- (A) The number of snakes will increase
- (B) The number of grasshoppers will increase and snakes will decrease
- (C) Nothing will change
- (D) The grass will disappear

Q10. We see different phases of the Moon, such as full moon and crescent. Why does the shape of the Moon appear to change?

- (A) The Moon makes its own light which grows and shrinks
- (B) Clouds cover different parts of the Moon
- (C) We see different amounts of the sunlit half of the Moon as it moves around the Earth
- (D) The Earth's shadow changes the Moon's size

Q11. Which group contains ONLY things that all living organisms need to survive?

- (A) Air, water, food, toys
- (B) Air, water, food, shelter
- (C) Water, food, mobile phone, air
- (D) Sunlight, television, water, air

Q12. A magnet is brought near a tray of mixed objects. Which set of objects will the magnet attract?

- (A) Iron pin, steel clip, plastic button
- (B) Iron pin, steel clip, nickel coin
- (C) Copper wire, wooden block, iron pin
- (D) Aluminium foil, plastic button, eraser

Q13. A tadpole slowly changes into a frog over several weeks — growing legs and losing its tail. This series of changes in an animal's life is called:

- (A) Migration
- (B) Hibernation
- (C) Metamorphosis
- (D) Germination

Q14. Wet clothes dry faster on a hot, windy day than on a cold, still day. Which process is making the clothes dry, and what speeds it up?

- (A) Condensation; speeded up by cold
- (B) Evaporation; speeded up by heat and wind
- (C) Freezing; speeded up by wind
- (D) Melting; speeded up by heat

Q15. Why is the Sun extremely important for almost all life on Earth? Choose the BEST reason.

- (A) It is the largest planet
- (B) It gives light and heat, and lets green plants make food, which feeds other living things
- (C) It is closest to the Moon
- (D) It only gives us day and night



Q16. Look at this set: cow, goat, deer, rabbit. Which statement is TRUE about all four animals?

- (A) They are all carnivores
- (B) They are all herbivores that eat plants
- (C) They all lay eggs
- (D) They all live in water

Q17. A doctor advises a child to eat more oranges, guavas and lemons to stay healthy and avoid getting sick easily. These foods are rich in:

- (A) Fats
- (B) Vitamin C
- (C) Roughage only
- (D) Proteins

Q18. An ice cube is left on a plate in a warm room. After an hour there is only a small puddle of water. Which TWO changes of state could happen next if the room stays warm all day?

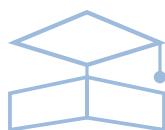
- (A) Melting, then evaporation
- (B) Freezing, then melting
- (C) Condensation, then freezing
- (D) Evaporation, then freezing

Q19. A bird has a strong, hooked beak and sharp claws called talons. From these features, what is this bird MOST likely to be?

- (A) A seed-eating bird like a sparrow
- (B) A bird of prey like an eagle that hunts other animals
- (C) A water bird that filters tiny plants
- (D) A bird that eats only nectar

Q20. Reasoning: 'Stethoscope is to Doctor as Chalk is to ____'. Which option completes the relationship in the same way?

- (A) Hospital
- (B) Teacher
- (C) Blackboard
- (D) Student



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	B	B	B	C	B	C	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	B	C	B	B	B	B	A	B	B

Step-by-Step Solutions

Q1. Correct answer: **(B)**

A potato is an underground **stem** (called a tuber). It is swollen because the plant stores extra food (starch) in it. The small 'eyes' on a potato are buds, which only stems have — roots do not have buds.

Q2. Correct answer: **(B)**

During photosynthesis, plants take in **carbon dioxide** and give out **oxygen**. Animals breathe in that oxygen — this is why plants and animals depend on each other.

Q3. Correct answer: **(B)**

A camel's hump stores **fat**, not water. When food and water are scarce, the body breaks this fat down to release energy and a small amount of water. A common myth is that the hump holds water directly.

Q4. Correct answer: **(B)**

Food goes: **mouth → food pipe (oesophagus) → stomach → small intestine → large intestine**. Digestion begins in the mouth (chewing and saliva), and most nutrients are absorbed in the small intestine.

Q5. Correct answer: **(B)**

The sugar has **dissolved** — its tiny particles spread out evenly in the water, so we cannot see them, but the sugar is still there (the water tastes sweet). Dissolving is a change you can reverse by evaporating the water.

Q6. Correct answer: **(C)**

Metals like **iron and copper** are good conductors, so electricity passes through them. Plastic is an insulator, so it does not let electricity pass — that is why wires are covered in plastic for safety.

Q7. Correct answer: **(B)**

Friction is the force between the ball and the ground that opposes motion and slowly stops the ball. On a smoother surface there is less friction, so the ball would roll farther before stopping.

Q8. Correct answer: **(C)**

Sunlight (solar energy) is renewable — it is available every day and will not run out in our lifetime. Coal, petrol and diesel are fossil fuels: they take millions of years to form and will eventually be used up.

Q9. Correct answer: **(B)**

Frogs eat grasshoppers, and snakes eat frogs. If frogs vanish, fewer grasshoppers are eaten, so **grasshoppers increase**; and snakes lose their food, so **snakes decrease**. This shows how every living thing in a food chain is connected.

**Q10. Correct answer: (C)**

The Moon has no light of its own; it reflects sunlight. As the Moon orbits the Earth, we see **different amounts of its sunlit half**, which makes it look like the shape is changing. (The Earth's shadow only causes an eclipse, not the regular phases.)

Q11. Correct answer: (B)

All living organisms need **air, water, food and shelter** to survive. Toys, mobile phones and televisions are human-made wants, not survival needs of living things.

Q12. Correct answer: (B)

Magnets attract magnetic materials such as **iron, steel and nickel**. They do not attract plastic, wood, copper or aluminium. Only option (B) contains three magnetic materials.

Q13. Correct answer: (C)

The change of form from a tadpole to an adult frog is called **metamorphosis**. (Germination is for seeds, migration is travelling to a new place, and hibernation is a long winter rest.)

Q14. Correct answer: (B)

The water in the clothes turns into water vapour by **evaporation**. **Heat and moving air (wind)** both make evaporation faster, so the clothes dry more quickly on a hot, windy day.

Q15. Correct answer: (B)

The Sun gives **light and heat**, and green plants use sunlight to make food. Since almost all animals depend on plants for food, the Sun is the original source of energy for nearly all life. (The Sun is a star, not a planet.)

Q16. Correct answer: (B)

A cow, goat, deer and rabbit all eat plants, so they are all **herbivores**. They give birth to young ones (they do not lay eggs) and live on land.

Q17. Correct answer: (B)

Citrus fruits like oranges, guavas and lemons are rich in **Vitamin C**, which helps the body fight illness and keeps gums healthy. Proteins are body-building foods found mostly in pulses, eggs and milk.

Q18. Correct answer: (A)

The solid ice first **melts** into liquid water. If the warm room continues, that water slowly turns into vapour by **evaporation**. So the order is melting → evaporation.

Q19. Correct answer: (B)

A strong **hooked beak** for tearing flesh and sharp **talons** for catching prey are features of a **bird of prey** such as an eagle or hawk. Seed-eaters have short cone-shaped beaks; nectar-feeders have thin long beaks. Body features tell us how an animal eats.

Q20. Correct answer: (B)

A doctor uses a stethoscope as a tool of their work; in the same way, a **teacher** uses chalk as a tool of their work. The relationship is 'tool → person who uses it', so the answer is teacher.